

Adam Speed-Andrews

Robotics & Mechatronics Engineer – MSc Electrical Engineering Candidate

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A Mechatronics Engineer with a passion for building robotic systems that operate reliably in complex real-world environments. My recent research with the African Robotics Unit focused on long-term autonomous navigation in agricultural settings, developing and evaluating visual systems capable of handling changing environmental conditions. Strong software foundations in Python and C++, with practical experience in system integration, field deployment, and experimental evaluation. Research published at ICRA 2026. Now based in London and eager to contribute to a team solving hard engineering problems – in robotics, computer vision, embedded systems, or beyond.

EDUCATION

University of Cape Town

Feb. 2024 – Present

MSc. Eng. Electrical Engineering (Research)

Cape Town, South Africa

- **Focus:** Long-term visual navigation of ground robots in dynamic agricultural environments using stereo vision.
- Supervised within the African Robotics Unit and Spatial Intelligence Lab.
- **Submitted** in April 2026, expected result in July 2026.

University of Cape Town

Feb. 2020 – Nov. 2023

BSc. Eng. Mechatronics (with Honours)

Cape Town, South Africa

- **Academic Standing:** First Class Equivalent (76% GPA)
- **Thesis:** Investigated the use of Neural Radiance Fields (NeRF) for appearance-based visual localization for robotics.
- Dean's Merit List 2021 & 2022. Golden Key International Honour Society member (2022)

WORK EXPERIENCE

African Robotics Unit (University of Cape Town)

Apr. 2025 — Present

Research Assistant – Robotics Test & Navigation

Cape Town, South Africa (Remote)

- Developed and deployed a novel language-enabled hierarchical navigation framework for agricultural robots, integrating natural language prompts into real-world mission planning.
- Co-authored two peer-reviewed publications; presented at ICRA 2026 in Vienna.

Dept. of Electrical Engineering (University of Cape Town)

Jul. 2024 – Nov. 2025

Teaching Assistant

Cape Town, South Africa

- Coordinated a 10-person tutor team providing academic support to 200+ students, including scheduling, creating teaching materials, and delivering occasional lectures.
- Provided structured feedback on technical assignments and practicals.

University of Cape Town

Feb. 2023 – Jun. 2024

University Tutor

Cape Town, South Africa

- Tutored undergraduate engineering students in Control Systems Engineering, Signals and Systems II, Electronic Devices and Circuits, and Analogue Electronics.
- Led weekly tutorial and practical sessions; responsible for marking and written feedback on assignments.

NUTEC Digital Ink

Jul. 2023 – Aug. 2023

Automation Design Trainee

Cape Town, South Africa

- Built and improved an adhesion testing machine to measure the tensile strength of laminate materials, incorporating smooth stepper motor control and simultaneous force data logging through an intuitive user interface.

PROJECTS & RESEARCH

Robust Long-Term Agricultural Robot Navigation

- Developed and evaluated vision-based teach and repeat navigation systems enabling robots to operate reliably across seasons and under significant lighting changes in agricultural crop row settings.
- Incorporated multi-experience route traversals to improve long-term robustness, validated through 10 km of autonomous crop-row navigation over a 6-month field trial.
- Stack: C++, Python, ROS2, OpenCV

Localization using Neural Radiance Fields (Honours Thesis)

- Researched Neural Radiance Fields for appearance-based visual localization, using novel-view synthesis to support place recognition in robotics.
- Benchmarked traditional visual bag-of-words methods against modern learned feature extractors on rendered NeRF imagery, demonstrating robust matching performance across synthesized novel views.
- Stack: Python, nerfstudio, PyTorch, OpenCV.

Automated Bird Weight Monitoring System

- Collaborated with the UCT FitzPatrick Institute of African Ornithology to design and build a non-invasive bird monitoring system for ecological research and health assessment.
- Designed and integrated a Raspberry Pi-based monitoring platform incorporating load cells, RFID identification, and cloud-based data storage to automatically collect and record bird weight measurements, enabling continuous monitoring of tagged birds.

PUBLICATIONS

Language Enabled Hierarchical Scene Graphs for Precision Agriculture Autonomy – ICRA 2026 Jun. 2026

Co-authored and presented at the IEEE International Conference on Robotics and Automation in Vienna on natural language-guided agricultural robotics, developing the navigation pipeline that integrated natural language prompts for autonomous robot navigation.

Osiris++: hierarchical representations for robotic-enabled precision agriculture – Frontiers Article Feb. 2026

Conducted field testing, data collection, and navigation performance evaluation for autonomous agricultural robot deployments, supporting the validation of the Osiris++ framework.

SKILLS

Programming: Python, C++, MATLAB, Java

Robotics & Perception: ROS/ROS2, Gazebo, RViZ, Foxglove, OpenCV, NeRFstudio

Hardware & EDA: KiCad, SolidWorks, LtSpice, 3D Printing, ESP32, STM32

Software & Dev Tools: Git, Linux (Ubuntu), Docker